

A Novel Three-Part Divisibility Test for 7

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1. Introduction

This document presents a new modular reduction formula for testing the divisibility of any integer by 7. Unlike standard methods that reduce numbers digit-by-digit, this approach uses a grouping method to achieve faster reduction.

2. The Formula

Let N be an integer. We define three variables based on the positions of its digits:

- y : The last two digits.
- x : The digit in the hundreds place.
- z : The remaining digits (thousands and above).

$$\text{Result} = ((5x + z) - y)$$

If the result is divisible by 7 (including 0 or negative multiples), then the original number N is divisible by 7.

3. Mathematical Proof

Any number N can be expressed as:

$$N = 1000z + 100x + y$$

Using modular arithmetic base 7:

- $1000 \equiv -1 \pmod{7}$
- $100 \equiv 2 \pmod{7}$

Thus: $N \equiv -z + 2x + y \pmod{7}$.

The Asaye formula $f = 5x + z - y$ satisfies:

$$-f = -5x - z + y \equiv 2x - z + y \pmod{7}$$

Since $-5 \equiv 2 \pmod{7}$, the formula is congruent to the original number's remainder.

4. Examples

Number (N)	x	z	y	Formula: (5x + z) - y	Result
1,652	6	1	52	(30 + 1) - 52 = -21	Divisible (Yes)
2,415	4	2	15	(20 + 2) - 15 = 7	Divisible (Yes)